Oct. ~~29~~ 30

Student Hours This Thursday moved from 5:45 pm start to 4:15 pm start still at MLC. Please come & work with classmates.

Those registered in MATH 440 can do project and we will use ~~that~~ the 340 scale to determine course grade.

Ch 8 Wave Eq'n

Section 1: Sample sol'ns

<u>Wave Eq'n</u> (hyperbolic eq'n)

$$\Delta u - u_{tt} = 0$$

sol'n: $u(x_1, x_2, \ldots, x_n; t)$ $\qquad (\underline{x}, t) \in \mathbb{R}^{n+1}$

spatial ↓ ↙ time

1-Dim wave eq'n: $x \in \mathbb{R}$ (still have 2 indep. variables)

2-Dim wave eq'n: $\underline{x} \in \mathbb{R}^2$ (3 indep. variables)

① Plane waves

Principle part $\Delta u - u_{tt} = 0$

charact. direction $\underline{\xi}$: $\underbrace{(\xi_1)^2 + (\xi_2)^2 + \ldots (\xi_n)^2 - (\xi_t)^2 = 0}_{\text{charact. eq'n}}$

direction is what is important (not magnitude)

Fix ξ_t, $|\xi_t| = 1$ i.e. $\xi_t = \pm 1$

For wave eq'n

$$\xi_1^2 + \xi_2^2 + \dots + \xi_n^2 = 1$$

<u>Def'n</u> The <u>characteristic surface</u> is a surface that has $\underline{\xi}$ as its normal.

Recall the eq'n of a plane

$$n_1 x_1 + n_2 x_2 + n_3 x_3 = d$$

$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0$

has normal vector $\langle n_1, n_2, n_3 \rangle$.

So $\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n - t = c$ ← a characteristic surface for $\Delta u - u_{tt} = 0$.

has normal vector $\langle \xi_1, \xi_2, \dots, \xi_n, -1 \rangle$

which is a charact. direction

So: The plane is a characteristic surface.

<u>claim</u> Plane wave sol'ns to the wave eqn are

$$u(x_1, x_2, \dots, x_n; t) = F(\underbrace{\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n - t}_{= c})$$

$$= F(c)$$

So while on plane, u is a constant.

<u>check</u> $\Delta u - u_{tt} = 0$?

$$u_{x_1} = F'(\qquad)\xi_1$$

$$u_{x_2} = F'(\qquad)\xi_2$$

$$\vdots$$

HW problem.

$\underline{n=1}$ $\qquad \underline{\xi} = \langle \xi_1, \pm 1 \rangle \qquad \xi_1^2 = 1$

2 charact. directions: $\langle 1, 1 \rangle$, $\langle 1, -1 \rangle$

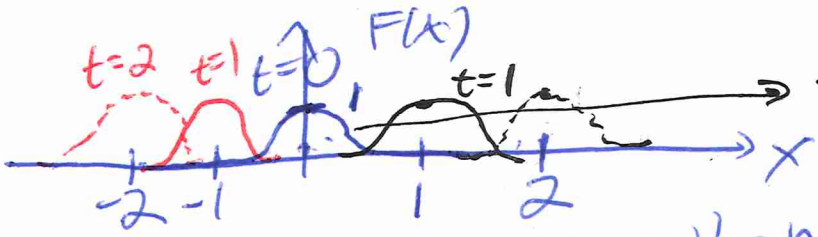
what about $\langle -1, -1 \rangle$?

Plane waves

$u(x,t) = F(x-t)$ moves to right

$u(x,t) = F(x+t)$ moves to left

Suppose $F(x)$ is give by a "blip"



to the right

The peak of $F(x)$ occurs when the argument is zero.

$F(x-1)$

$F(x-t)$: $t=1$ peak occurs $x=1$

$t=2$ peak occurs $x=2$

$F(x+t)$: $t=1$ peak occurs $x=-1$

$t=2$ " " $x=-2$

Recall, ch5 section 7, The sol'n to

$u_{xx} - u_{tt} = 0 \iff u_{\xi\eta} = 0 \qquad \xi = x+t, \quad \eta = x-t$

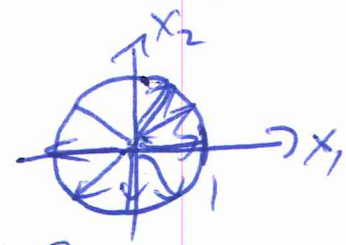
solved by integrating twice:

$u(x,t) = F(x+t) + G(x-t)$ (general sol'n - all sol'ns have this form!)

$\underline{n=2}$ $\quad \xi_1^2 + \xi_2^2 = 1 \qquad \underline{\xi}_x =$

# of characteristic directions: $\langle 0,1\rangle, \langle 1,0\rangle, \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\rangle$

∞-many

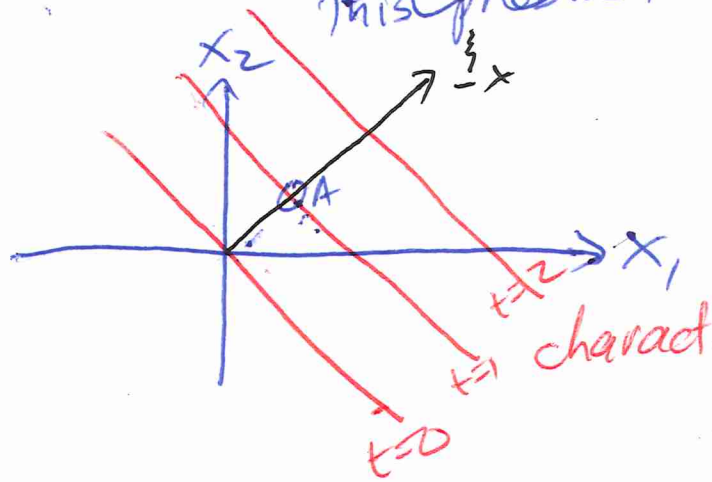


any vector that has magnitude 1.

ex)

$$u(x_1,x_2,t) = F\left(\tfrac{1}{\sqrt{2}}x_1 + \tfrac{1}{\sqrt{2}}x_2 - t\right)$$

$$\underline{\xi}_x = \left\langle \tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}} \right\rangle \qquad \underbrace{\underline{\xi} = \left\langle \tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, 1 \right\rangle}_{\text{This problem}}, \qquad \underbrace{\left\langle \tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}, -1 \right\rangle}_{\text{This problem}}$$



charact. "surfaces"

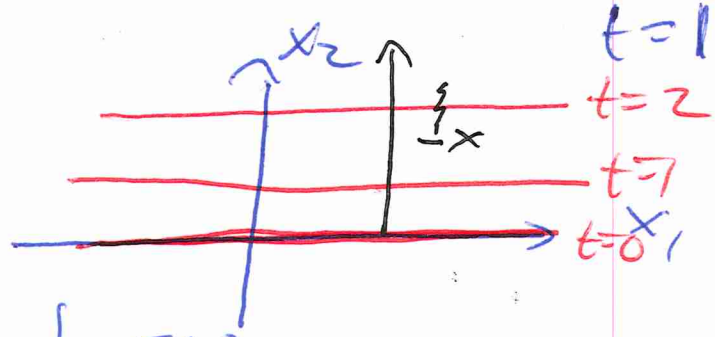
$A = \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right)$

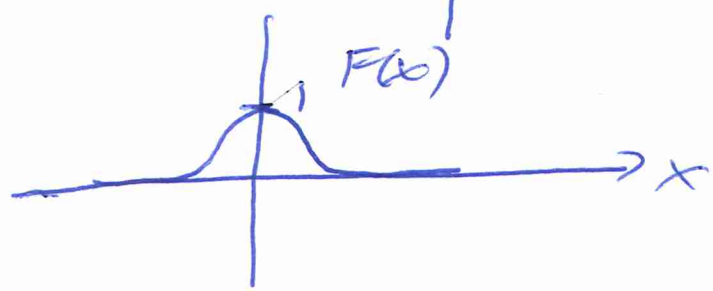
$$\left(\tfrac{1}{\sqrt{2}}\right)\left(\tfrac{1}{\sqrt{2}}\right) + \left(\tfrac{1}{\sqrt{2}}\right)\left(\tfrac{1}{\sqrt{2}}\right) - t = 0$$

Another choice: $\underline{\xi}_x = \langle 0, 1 \rangle$

$$u(x_1,x_2,t) = F(x_2 - t)$$

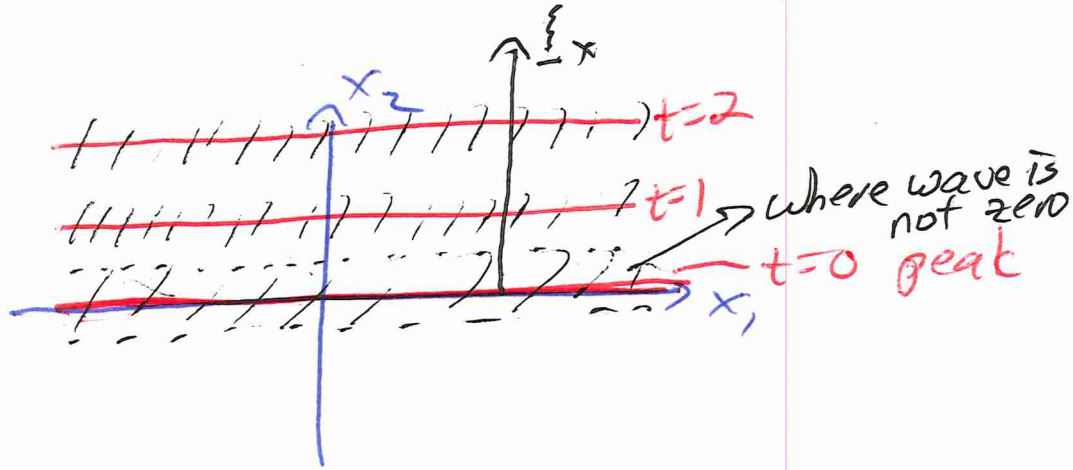
Suppose $F(x)$





blip fn

Then the peak occurs when the argument of F is zero.



$u(x_1, x_2, t) = F(x_2 - t)$

$t=0$ peak occurs $x_2 = 0$

$t=0$ $x_2 = \pm 1$

blip is zero when argument is ± 1

$t=1$ peak occurs $x_2 = 1$

$t=2$ peak occurs $x_2 = 2$

② Spherical sol'ns (sol'ns that do not depend on θ, ϕ)

wave eq'n (n, spherical)

def'n of spherical sol'n

$$\Delta u - u_{tt} = 0 = \frac{1}{r^{n-1}} \frac{\partial}{\partial r}\left(r^{n-1} \frac{\partial u}{\partial r}\right) + \frac{1}{r^2} \cancelto{0}{\Delta_n u} - u_{tt} = 0$$

check. ok ✓

<u>$n=3$</u>

$$\frac{1}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) - u_{tt} = 0$$

<u>Trick</u> $w = ru$ (change dependent variable)

$$\Rightarrow u = \frac{w}{r}$$

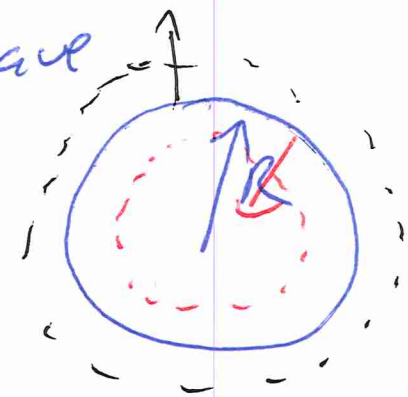
$$\frac{\partial u}{\partial r} = \frac{\partial}{\partial r}\left(\frac{w}{r}\right) = \frac{1}{r} w_r - \frac{1}{r^2} w \quad \text{etc.}$$

we get

$$w_{rr} - w_{tt} = 0 \quad \text{(wave eq'n in 1-D)!}$$

$$w(r,t) = F(r-t) + G(r+t)$$ (1 Dim general soln)

$$u(r,t) = \underbrace{\frac{F(r-t)}{r}}_{\text{outgoing wave}} + \underbrace{\frac{G(r+t)}{r}}_{\text{incoming wave}}$$



③ <u>What about separation of variables?</u>

(non-spherical sol'ns)

$u(\underline{x},t) = V(\underline{x})\,T(t)$

Following the same procedure as before, we get

$\Delta V + \mu V = 0$ ← <u>elliptic wave eq'n</u>, also known as <u>the reduced wave eq'n</u>

$T'' + \mu T = 0$ If $\mu > 0$, say $\mu = \omega^2$, then we get $\cos(\omega t)$, $\sin(\omega t)$

To solve $\Delta V + \mu V = 0$

could separate again: $V = R(r)\, Y(\theta, \phi)$

The sol'n Y_k = Laplace spherical harmonics

$R_{k+\frac{1}{2}}$ = Bessell Functions J_ν, I_ν